

AIDAN COOLEY

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EDUCATION

Bachelor of Science: Computer Science, 3.83 GPA

University of Utah, Kahlert School of Computing
Minor in Applied Mathematics

Salt Lake City, UT
Anticipated December 2026

Related Coursework

Algorithms, Computer Systems, Operating Systems, Data Wrangling, Computer Organization, Models of Computation, Real Analysis, Linear Algebra, ODEs

TECHNICAL SKILLS

Programming Languages: C, VHDL, SystemVerilog, Python, C++, Java, R, C#, SQL, QT

Technical Skills: Github, Vivado, Embedded Design, Firmware Design, Computer Engineering and Architecture, Scrum

EXPERIENCE

Los Alamos National Lab

CAI-3 Intern

Los Alamos, NM
May 2024 – Present

- Low level computer engineering for space missions
- Design efficient algorithms for satellite modelling and scheduling
- Work on a large team of engineers for products close to launch day
- Work on microcontrollers, FPGAs, and models using C, C++, Python, and VHDL

Wasatch Magazine

Head Editor

Salt Lake City, UT
May 2025 – May 2026

- Run a team of more than a dozen contributors
- Oversee the entire lifecycle of content from pitch to publish.
- Edit all the content written for the magazine
- Work alongside professionals in the industry on advertising, printing, distribution, and collaboration
- Write, take photos, and make videos for our publication

Publications and Presentations

- Aidan Cooley, "Spacewire on the Raspberry Pi Pico," poster presented at the LANL Student Symposium Aug. 2024, LA-UR-24-26829
- Aidan Cooley, "Debugging FPGAs Through a Software Slowdown," talk given at internal LANL student presentation Aug. 2025

TECHNICAL PROJECTS

Debugging FPGAs Through a Software Slowdown – Skills used: VHDL, C

May 2025 – Aug 2025

- Develop a test suite for a spacewire router on an FPGA for satellites featuring:
 - Fully customizable data packets, router paths, and speeds
 - Comprehensive error checking through log files
 - User friendly shell allowed for engineers to communicate with the router

Spacewire on the Raspberry Pi Pico – Skills used: C, microprocessor assembly

May 2024 – Aug 2024

- Implement a system to simulate behavior of spacewire on a \$4 microprocessor
- Utilize every piece of microprocessor to run a very complex data transfer protocol
- Showcases lowest bound of needed hardware to implement complex protocol
- Implemented SpaceWire on two \$4 microcontrollers, replacing FPGA test hardware costing thousands.

Assessing ML Models' Carbon Footprint on an FPGA – Skills used: HPC, Verilog, Pytorch

Sep 2025 – Dec 2025

- Government funded R&D for mission critical deterrence
- Develop algorithms and data structures to model satellite capabilities
- Implement fast validation schemes to pair with real time scheduling

VOLUNTEER WORK

Utah Climbing Team

Salt Lake City, UT

Outdoor Trip Manager

August 2023 – May 2026

- Plan and lead all the outdoor trips for the climbing team
- Teach outdoor climbing technique and safety

BSA Eagle Scout

Received July 2022